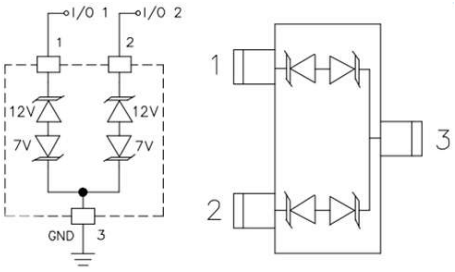
SM712

Asymmetrical TVS Diode Array

FEATURES:

- ✧ Ultra low leakage: nA level
- ✧ Operating voltage: 7V or 12V
- ✧ Low clamping voltage
- ✧ Complies with following standards:
  - – IEC 61000-4-2 (ESD) immunity test
    - Air discharge: ±15kV
    - Contact discharge: ±8kV
  - – IEC61000-4-5 (Lightning) 13A for 12V (8/20μs)
  - – IEC61000-4-5 (Lightning) 18A for 7V (8/20μs)
- ✧ RoHS Compliant

SOT-23



PIN Configuration

MAIN APPLICATIONS

- ✧ Protection of RS-485 transceivers with extended common-mode range
- ✧ Security systems
- ✧ Automatic Teller Machines
- ✧ HFC systems
- ✧ Networks

MECHANICAL CHARACTERISTICS

- ✧ Package SOT-23
- ✧ Molding Compound Flammability Rating : UL 94V-O
- ✧ Quantity Per Reel : 3,000pcs
- ✧ Lead Finish : Lead Free
- ✧ Marking : 712

ABSOLUTE MAXIMUM RATINGS (T_A=25°C, RH=45%-75%, unless otherwise noted)

Parameter	Symbol	Value	Unit
Peak Pulse Power (tp = 8/20μs),(V_{RWM}=7V/V_{RWM}=12V)	P_{PPM}	360	W
Peak Pulse Current (tp = 8/20μs),(V_{RWM}=7V/V_{RWM}=12V)	I_{PPM}	18/13	A
ESD voltage IEC 61000-4-2 (air discharge)	V_{ESD}	15	kV
ESD voltage IEC 61000-4-2 (contact discharge)	V_{ESD}	8	kV
Maximum lead temperature for soldering during 10s	T_L	260	°C
Storage Temperature Range	T_{stg}	-55 to +150	°C
Operating Temperature Range	T_{OP}	-55 to +125	°C

Electrical Characteristics(TA=25°C unless otherwise specified)

Parameter	Symbol	Min	Typ	Max	Unit	Condition
Reverse Working Voltage	V_{RWM}	--	--	7.0	V	Pin3 to Pin1 or 2
		--	--	12.0		Pin1 or 2 to Pin3
Breakdown Voltage	V_{BR}	7.5	--	--	V	I_T=1mA,Pin3 to Pin1 or 2
		13.3	--	--		I_T=1mA,Pin1 or 2 to Pin3
Leakage Current ILeak	I_R	--	--	1.0	μA	V_{RWM}=7V,Pin3 to Pin1 or 2
		--	--	1.0		V_{RWM}=12V,Pin1 or 2 to Pin3
Clamping Voltage	V_C	--	--	20.0	V	I_{PP}=18A,T_p=8/20μs,Pin3 to Pin1 or 2
		--	--	30.0		I_{PP}=13A,T_p=8/20μs,Pin1 or 2 to Pin3
Junction Capacitance	C_J	--	45.0	--	pF	V_R=0V, f=1MHz,Pin1 or 2 to Pin3

Characteristic Curves

FIG.1:V- I curve characteristics (Bi-directional)

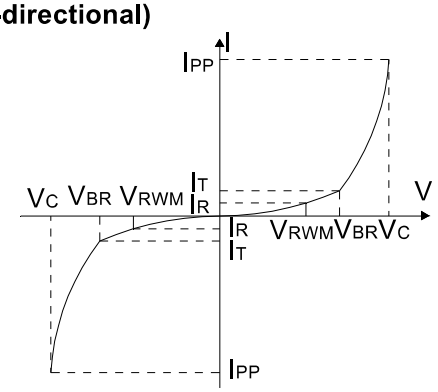


FIG.2: Pulse waveform (8/20μs)

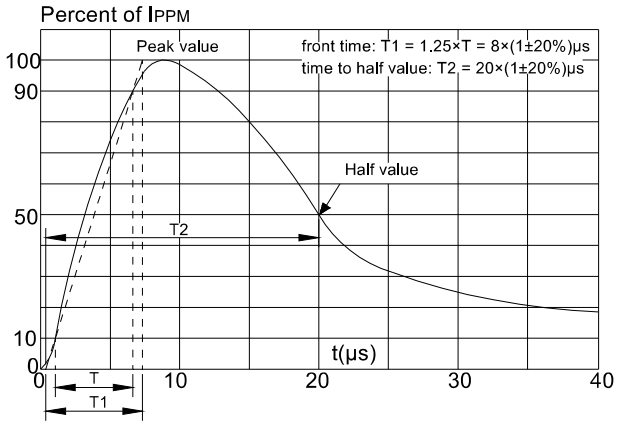


FIG.3: Pulse derating curve

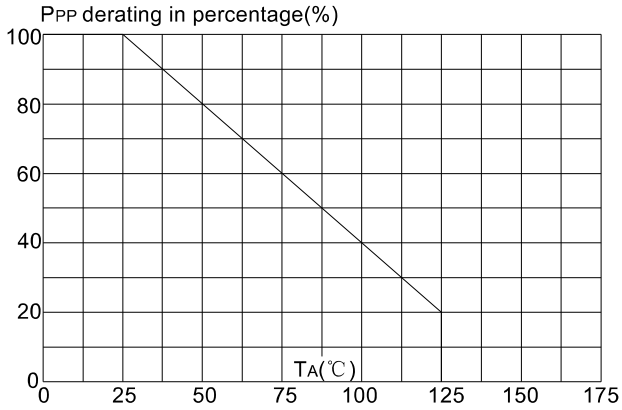
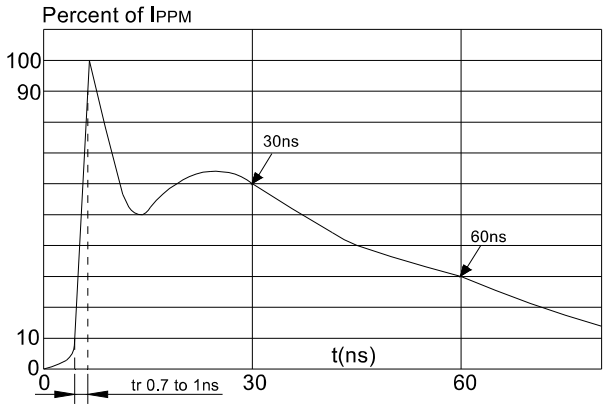
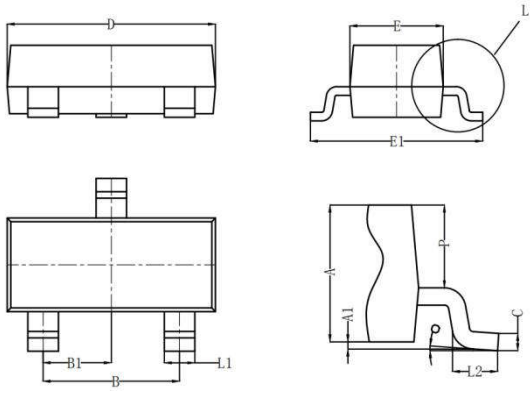


FIG.4: ESD clamping (8KV contact)



PACKAGE MECHANICAL DATA

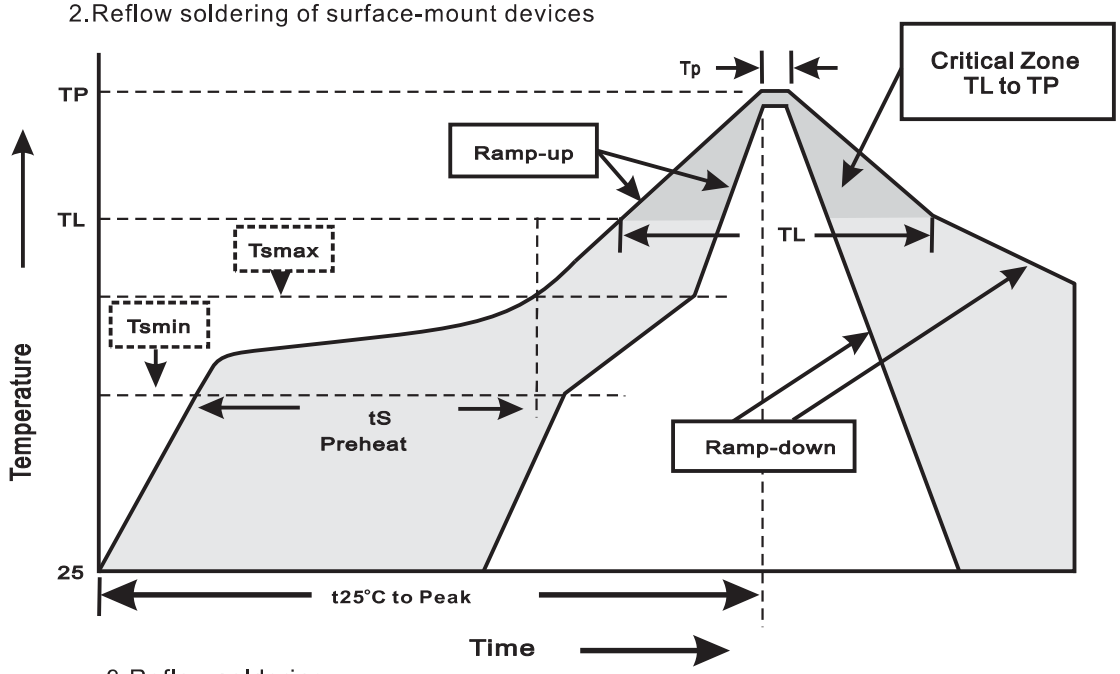


SOT-23

Symbol	Dimensions (mm)		
	Min	Typ	Max
A	0.900	1.000	1.1100
A1	0.000	0.050	0.100
L1	0.350	0.400	0.500
C	0.100	0.110	0.120
D	2.800	2.900	3.000
E	1.250	1.300	1.350
E1	2.250	2.400	2.550
B	1.800	1.900	2.000
B1	0.950 Typ		
L2	0.200	0.350	0.450
P	0.550	0.575	0.600

Suggested thermal profiles for soldering processes

1.Storage environment: Temperature=5°C~40°C Humidity=55%±25%
2.Reflow soldering of surface-mount devices



3.Reflow soldering

Profile Feature	Soldering Condition
Average ramp-up rate(T_L to T_P)	<3°C/sec
Preheat -Temperature Min(Tsmin) -Temperature Max(Tsmax) -Time(min to max)(t_s)	 150°C 200°C 60~120sec
Tsmax to T_L -Ramp-upRate	 <3°C/sec
Time maintained above: -Temperature(T_L) -Time(t_L)	 217°C 60~260sec
Peak Temperature(T_P)	255°C-0/+5°C
Time within 5°C of actual Peak Temperature(t_P)	10~30sec
Ramp-down Rate	<6°C/sec
Time 25°C to Peak Temperature	<6minutes